

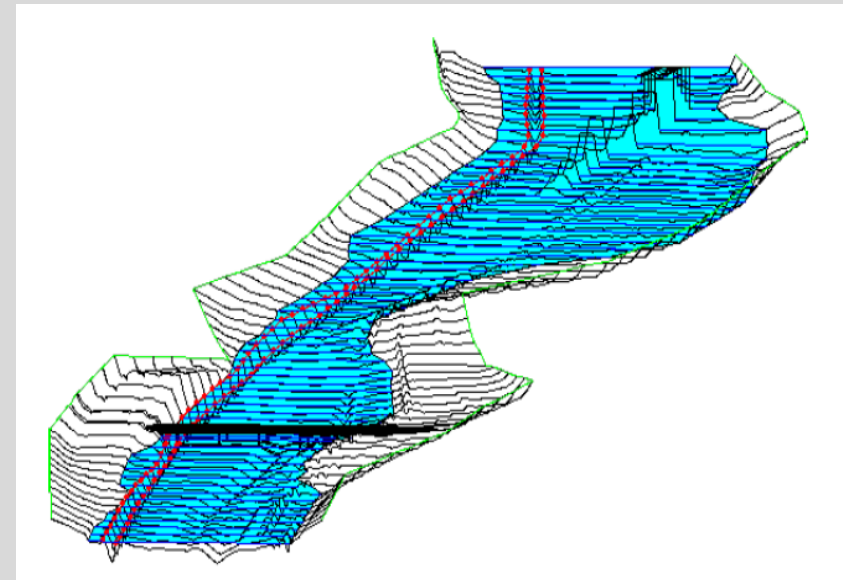
ENCN442 Surface Water and Groundwater Engineering



From Water Archives. More on the Teton Dam failure:

https://en.wikipedia.org/wiki/Teton_Dam

Surface Water Modelling
Hydraulic Modelling
- unsteady -



Learning Objectives (check later...)

After completion of this module the learner will be able to

- Explain the difference between steady and unsteady hydraulic modelling.
- Describe for which engineering problems unsteady hydraulic modelling is required.
- Discuss the different terms of the momentum equation and their magnitude.
- Describe different approaches to model flow through a culvert or bridge.
- Set up and use a 1D hydraulic model to assess the effect of constructing a culvert or bridge on water levels.

Materials for the module on hydraulic modelling

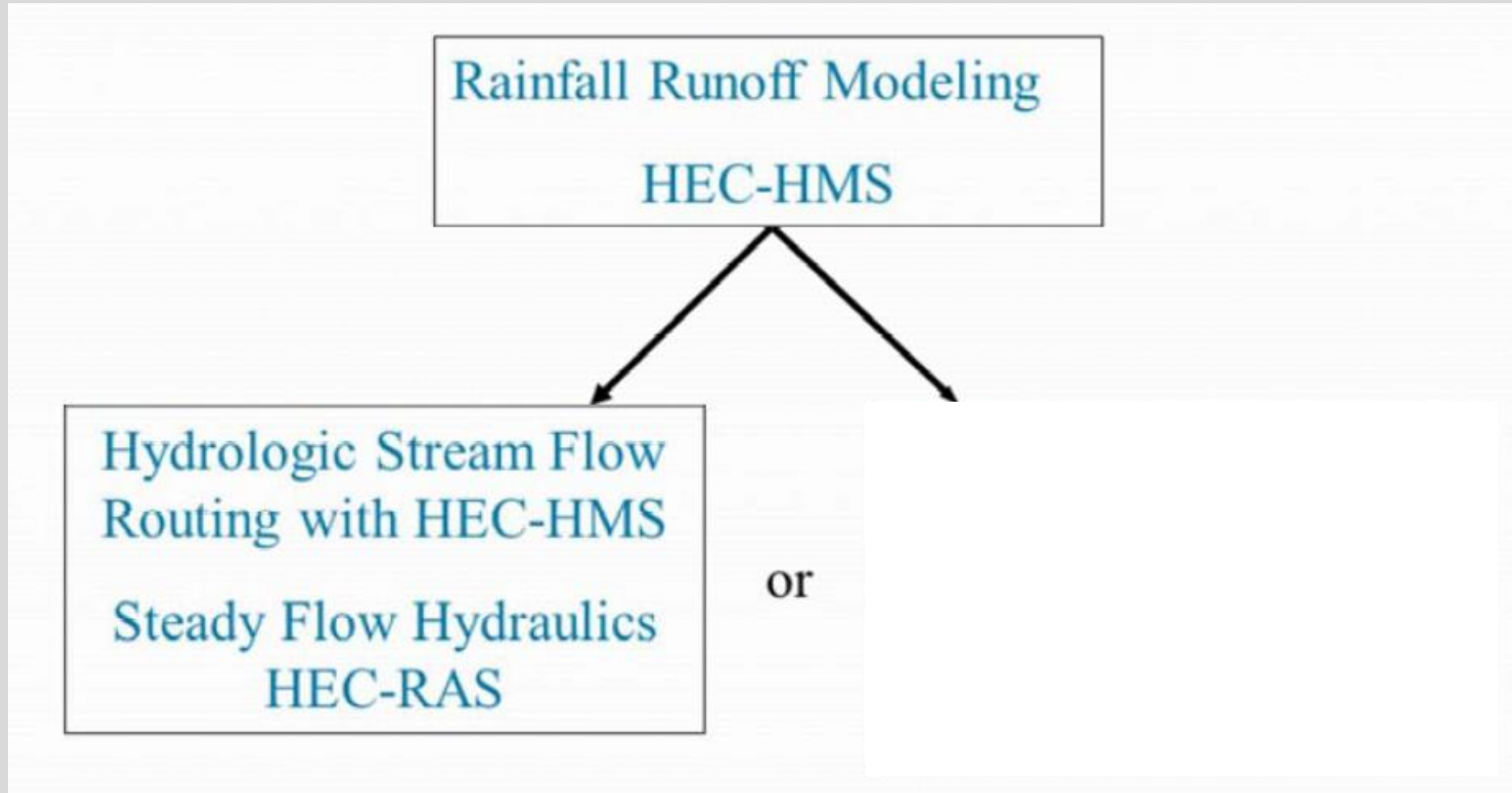
Course reader (available on LEARN).

Lecture slides and notes (available on LEARN before/after).

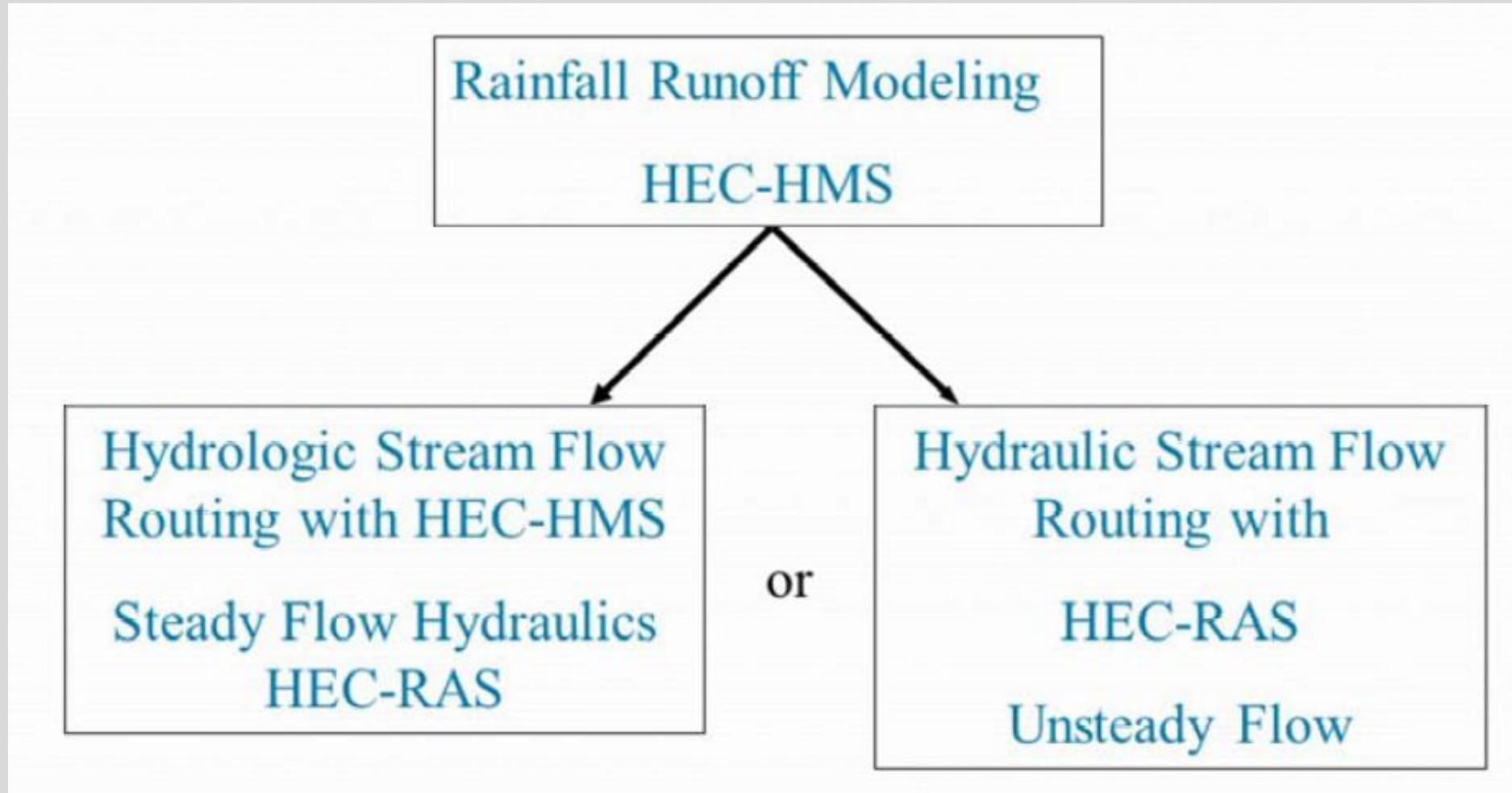
HEC-RAS software and documentation, available at <https://www.hec.usace.army.mil/software/hec-ras/> (open access).

Installed on computers in various UC computer labs (can be installed via UC 'Software Centre' if not already installed).

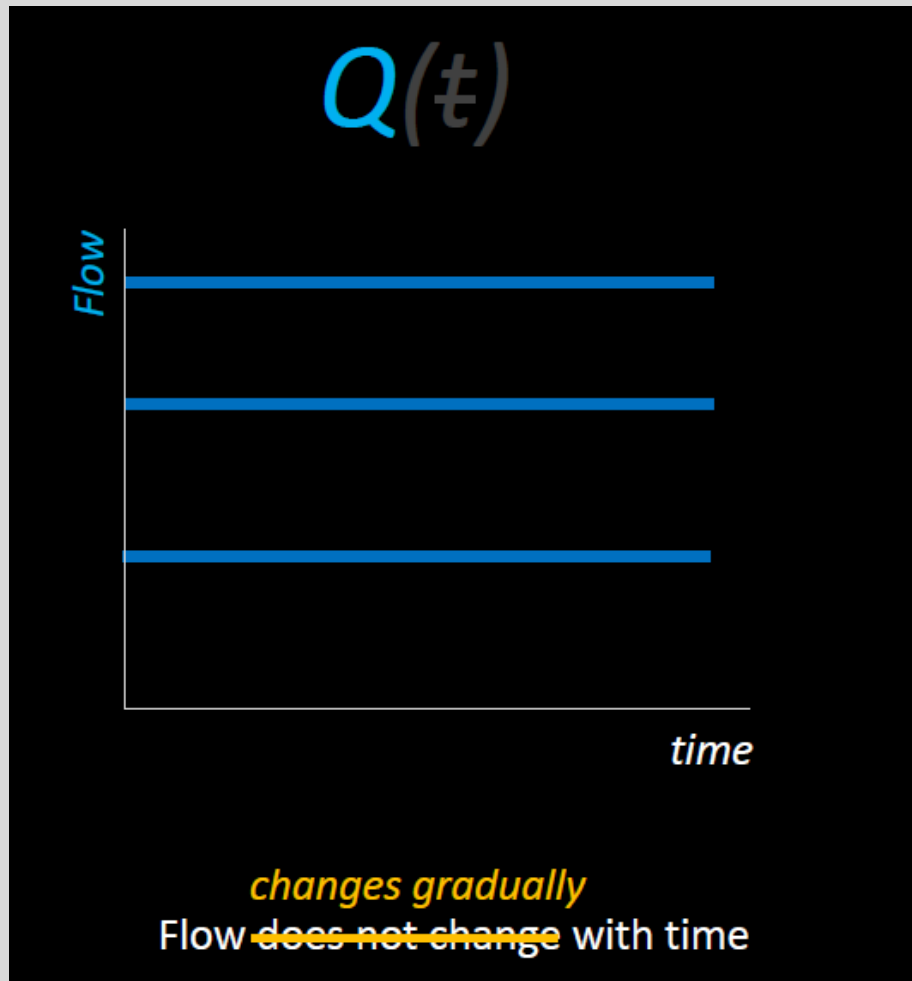
Stream Flow Routing Options



Stream Flow Routing Options



Recall: Steady vs Unsteady Flow



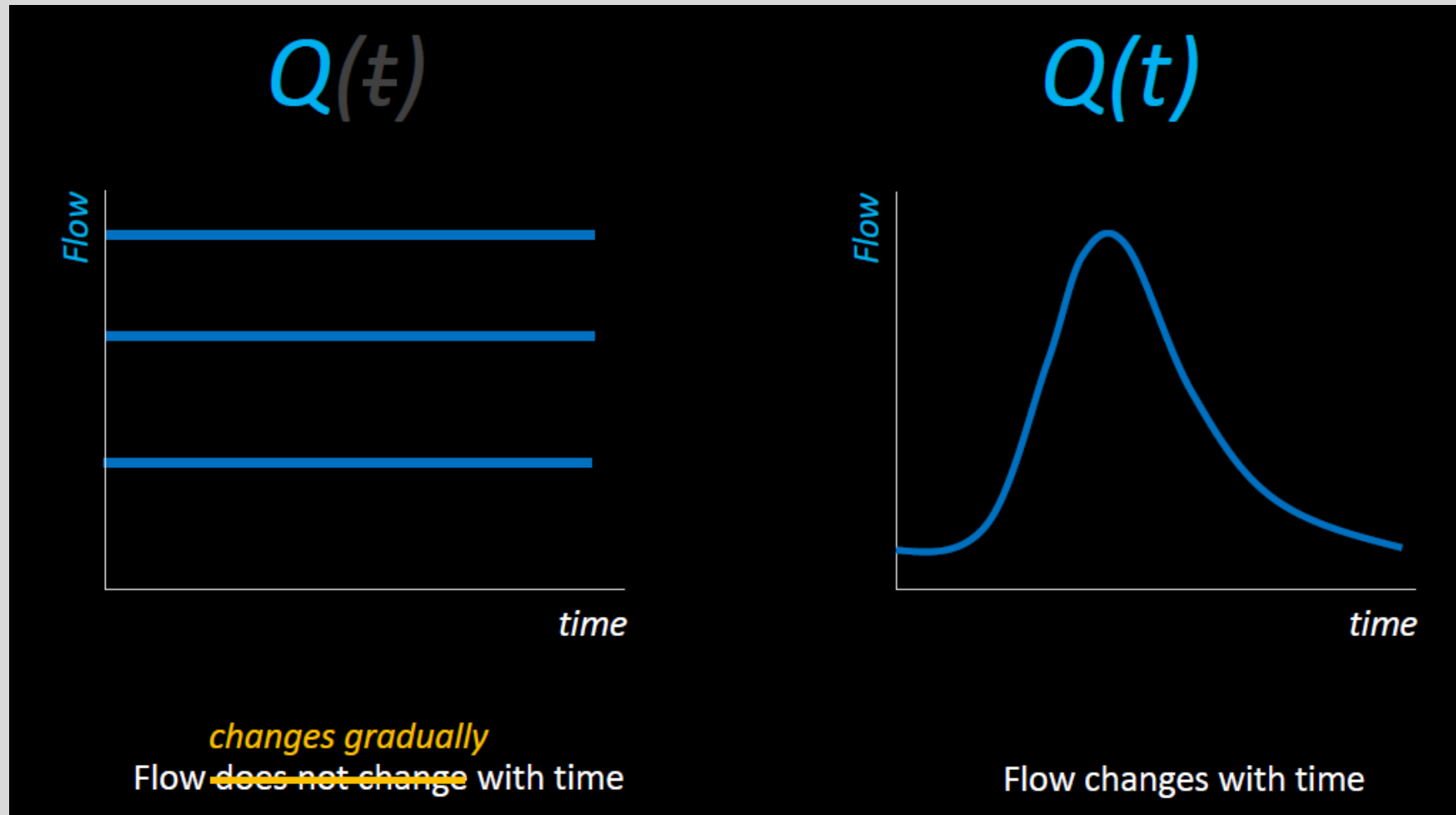
Steady Flow

- **Definition:** Flow (i.e. Depth, V , Q) does not change with time.
 - More realistically: Flow changes gradually with time, such that a steady flow assumption is reasonable for hydraulic calculations.

Steady Flow

- **Definition:** Flow (i.e. Depth, V , Q) does not change with time.
 - More realistically: Flow changes gradually with time, such that a steady flow assumption is reasonable for hydraulic calculations.
- **Examples:**
 - Flow in a water distribution canal.
 - Natural Floods from rainfall runoff events
 - Flow changes gradually
 - Modeling shorter distances where they peak flow is relatively steady over the shorter distance.

Recall: Steady vs Unsteady Flow



Unsteady Flow

- **Definition:** Flow (i.e. Depth, V , Q) changes with time.

Unsteady Flow

- **Definition:** Flow (i.e. Depth, V , Q) changes with time.
- **Examples:**
 - Dam Break
 - Tidal Effected Streams
 - Power Releases at reservoirs
 - Flash Floods
 - Tributary flow reversals due to backwater
 - Natural Floods from rainfall runoff events

Unsteady Flow Equations (Saint-Venant Equations)

Momentum Equation:

$$\frac{\partial Q}{\partial t} + \frac{\partial(\alpha Q^2/A)}{\partial x} + gA\left(\frac{\partial h}{\partial x} - S_o + S_f\right) = 0$$

Continuity Equation:

$$\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0$$

Unsteady Flow Equations – remove time-dependent terms to obtain

Momentum Equation:

$$\cancel{\frac{\partial Q}{\partial t}} + \frac{\partial(\alpha Q^2/A)}{\partial x} + gA\left(\frac{\partial h}{\partial x} - S_o + S_f\right) = 0$$

Continuity Equation:

$$\frac{\partial Q}{\partial x} + \cancel{\frac{\partial A}{\partial t}} = 0$$

Steady Flow Equations

Steady Flow form of Momentum Equation:

$$\frac{\partial(\alpha Q^2/A)}{\partial x} + gA\left(\frac{\partial h}{\partial x} - S_o + S_f\right) = 0$$

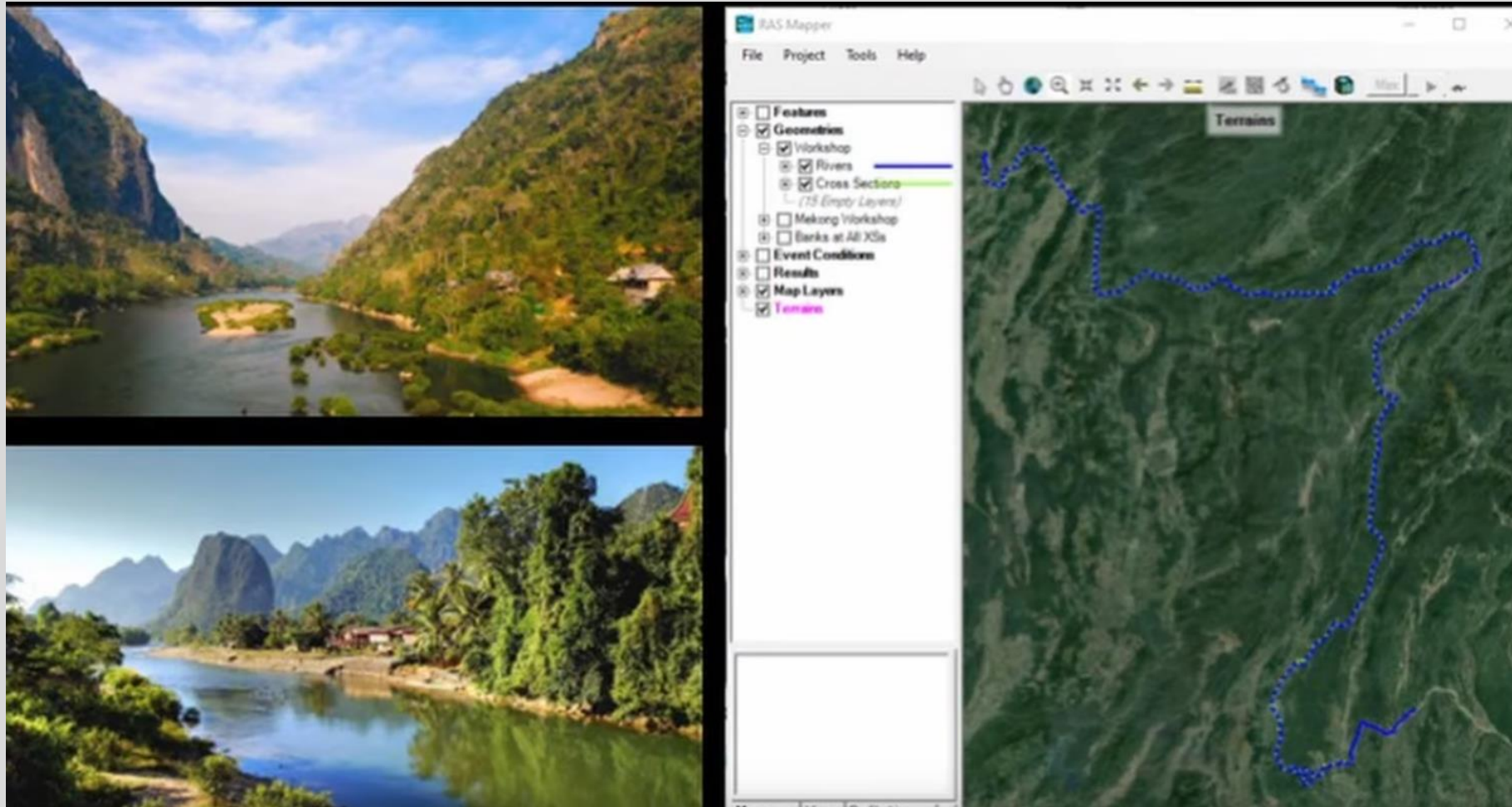
Continuity Equation:

$$Q = VA$$

Hydrologic vs Hydraulic Routing

- **Hydrologic routing** consists of the continuity equation and a relationship between storage in the reach and discharge at the outlet.

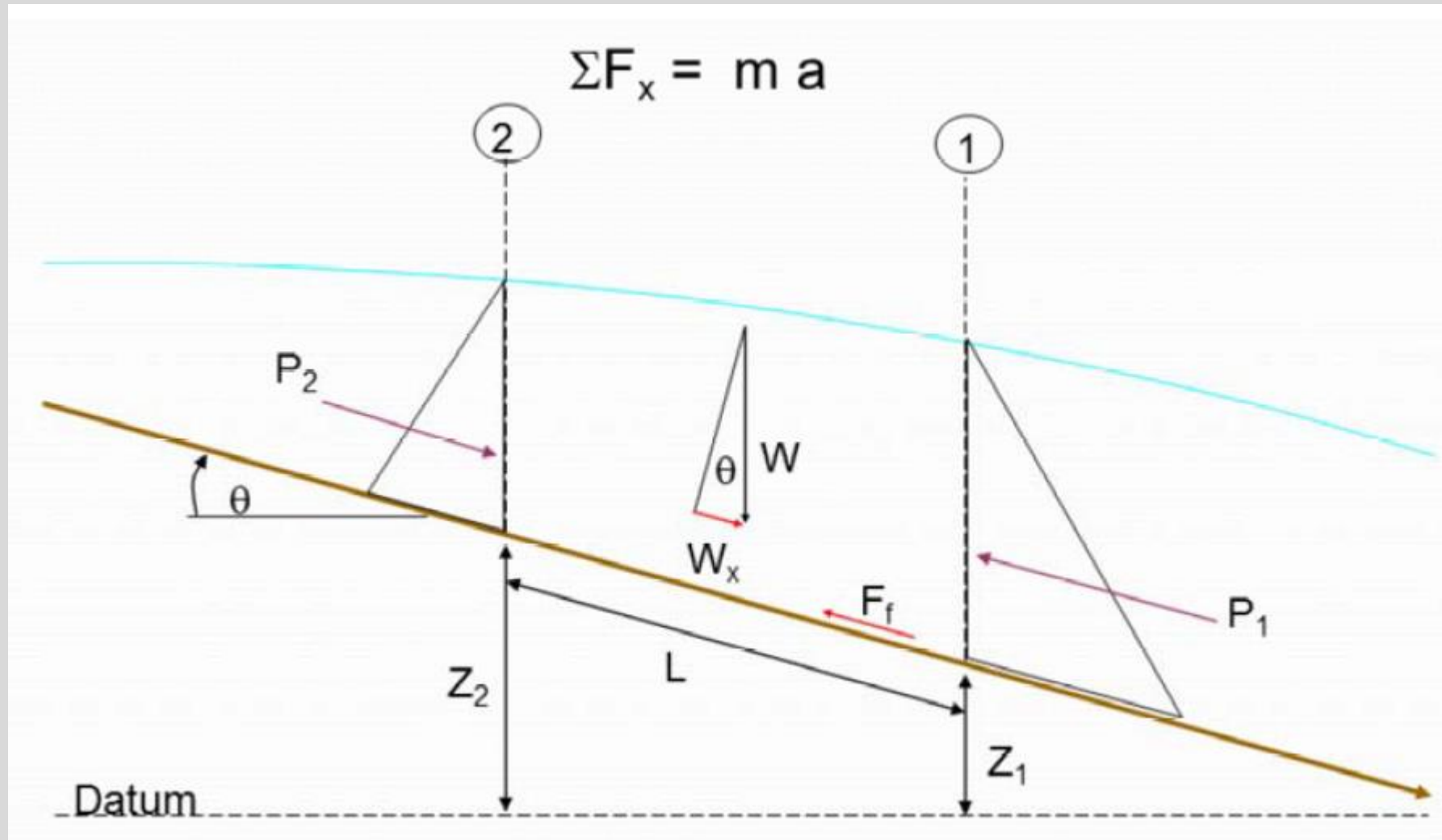
Quick check: steady or unsteady?



Hydrologic vs Hydraulic Routing

- **Hydrologic routing** consists of the continuity equation and a relationship between storage in the reach and discharge at the outlet.
- **Hydraulic routing** (Unsteady Flow) consists of the continuity equation and the momentum equation. Unsteady Flow routing is more physically based, and is applicable to a wider range of applications.

Momentum Equation



Momentum Equation Terms

$$S_f = S_o - \frac{\partial y}{\partial x} - \frac{V}{g} \frac{\partial V}{\partial x} - \frac{1}{g} \frac{\partial V}{\partial t}$$

- S_f • Friction slope (frictional forces)
- S_o • Bed slope (gravitational effects)

- $\frac{\partial y}{\partial x}$ • Pressure differential

- $\frac{V}{g} \frac{\partial V}{\partial x}$ • Convective acceleration

- $\frac{1}{g} \frac{\partial V}{\partial t}$ • Local acceleration

Approximations

$$S_f = S_o - \frac{\partial y}{\partial x} - \frac{V}{g} \frac{\partial V}{\partial x} - \frac{1}{g} \frac{\partial V}{\partial t}$$

- Steady Uniform Flow

Kinematic Wave Approximation

- Steady Nonuniform Flow

Diffusive Wave Approximation

- Steady Nonuniform Flow

Quasi-Steady State Dynamic Wave Approximation

- Unsteady Nonuniform Flow

Full Dynamic Wave Equation

Magnitudes of different Terms in the Momentum Equation (steep river!)

• Term	S_0	$\frac{\partial y}{\partial x}$	$\frac{V}{g} \frac{\partial V}{\partial x}$	$\frac{1}{g} \frac{\partial V}{\partial t}$
• Magnitude	26	.5	.12-.25	.05

Figures from Henderson 1966, with a hydrograph that rises and decreases in 24 hours from 10,000 cfs to 150,000 cfs to 10,000 cfs

- With smaller bed slopes, the inertial terms become more relevant

Unsteady Flow: How is it different than Steady Flow?

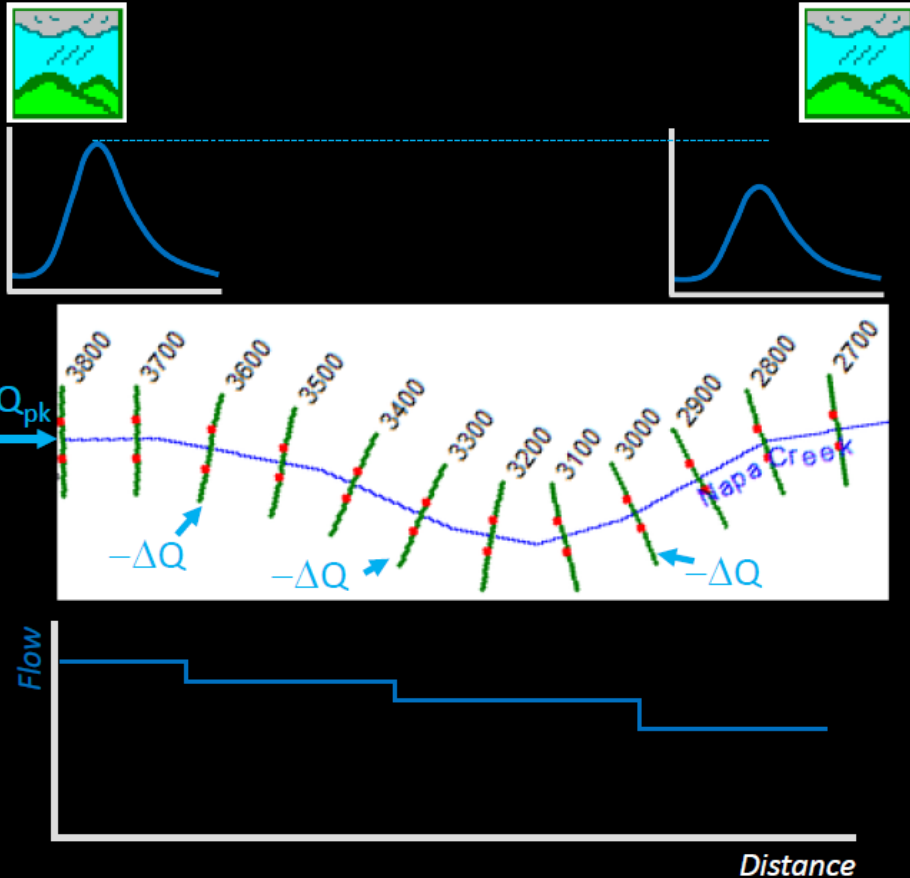
- Boundary Conditions
- Calibration
- Model Applications

Unsteady Flow: How is it different than Steady Flow?

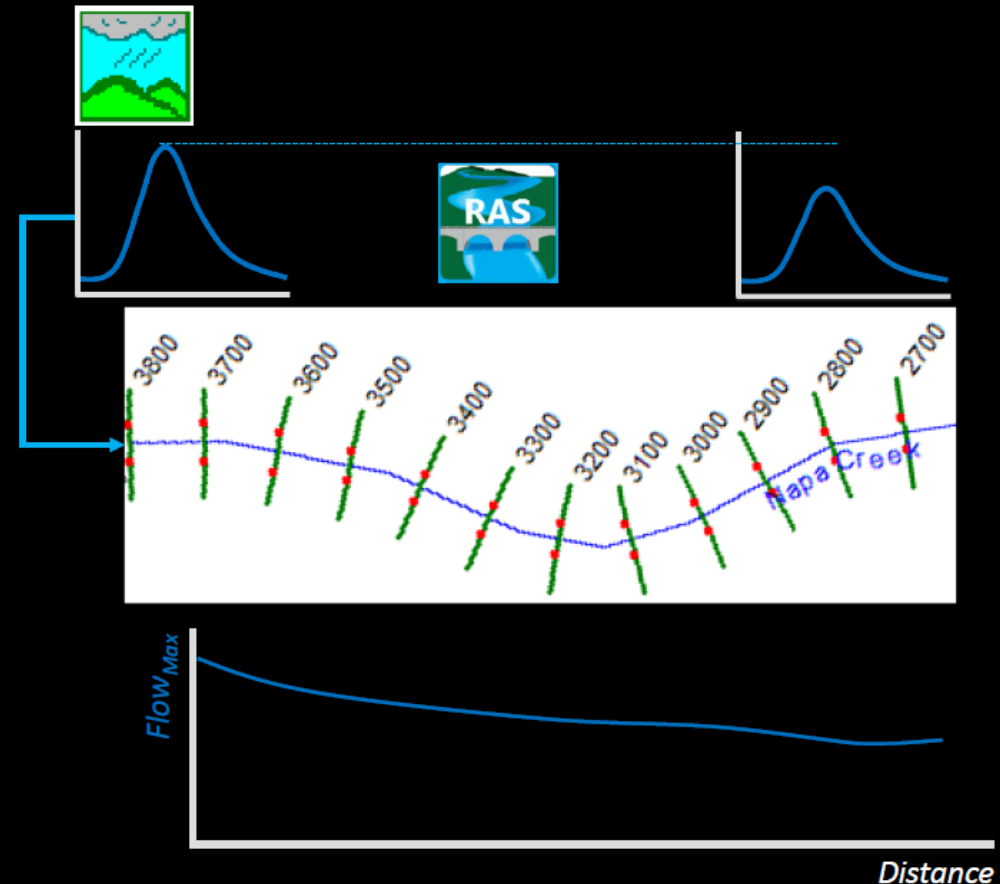
- **Boundary Conditions**
- Calibration
- Model Applications

Example: Flood Attenuation

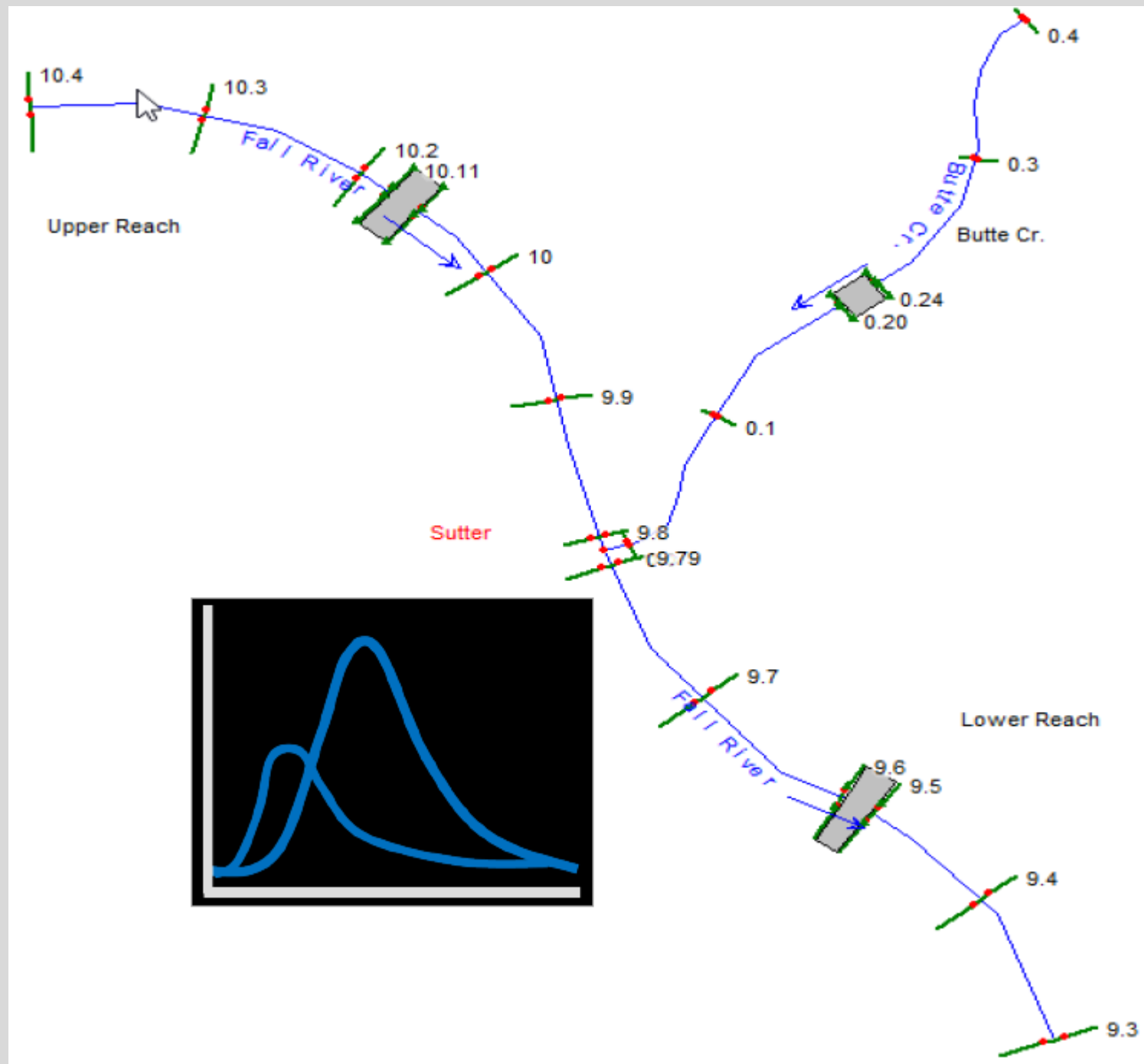
Hydrologic Routing Steady Flow



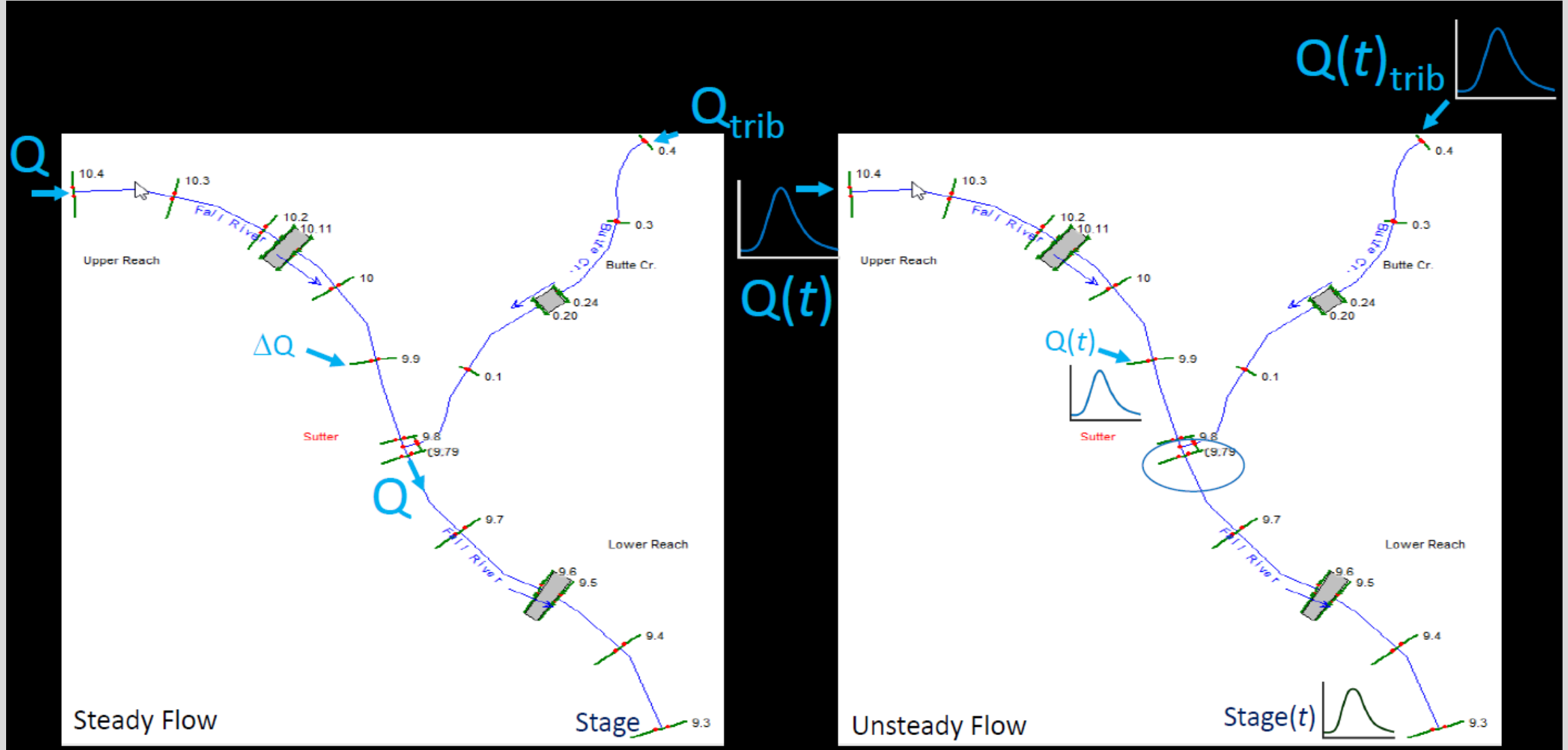
Hydraulic Routing Unsteady Flow



Example: Tributary Timing



Example: Tributary Timing



Unsteady Flow: How is it different than Steady Flow?

- Boundary Conditions
- **Calibration**
- Model Applications

Calibration

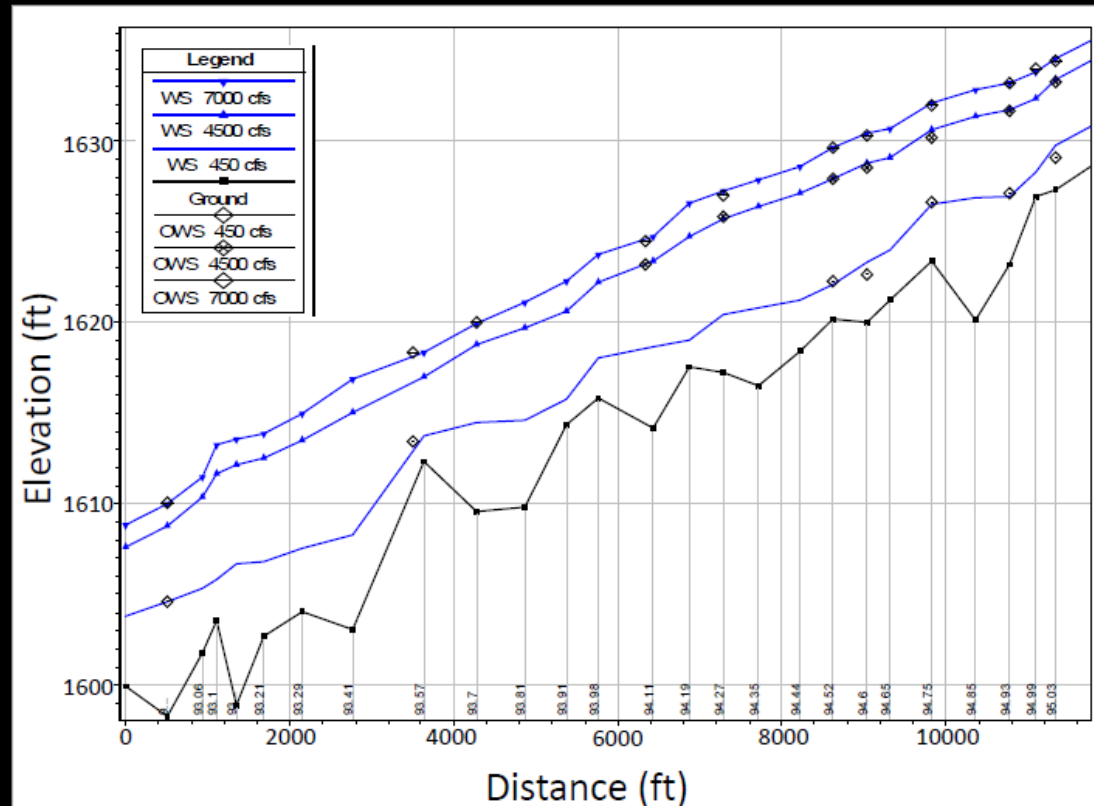


Hydraulic Modelling



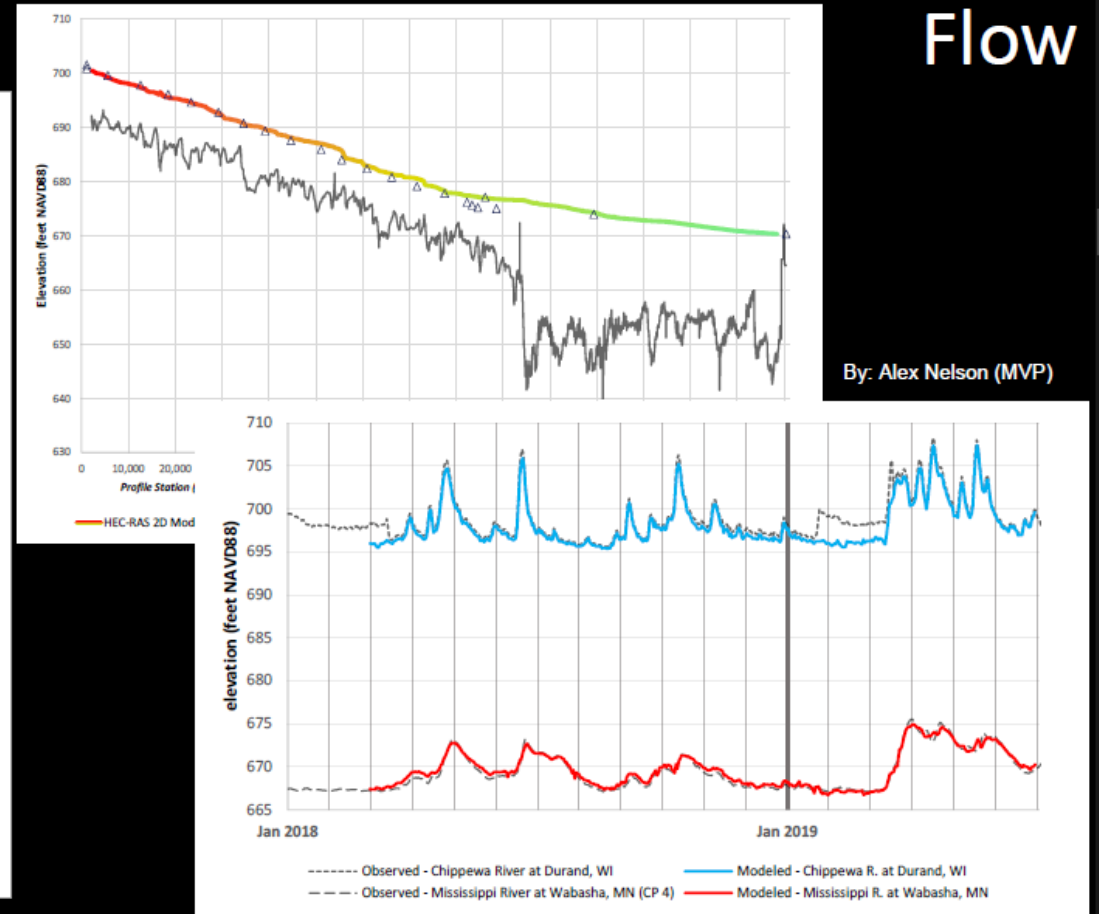
Calibration

Steady Flow



Calibration

Unsteady Flow

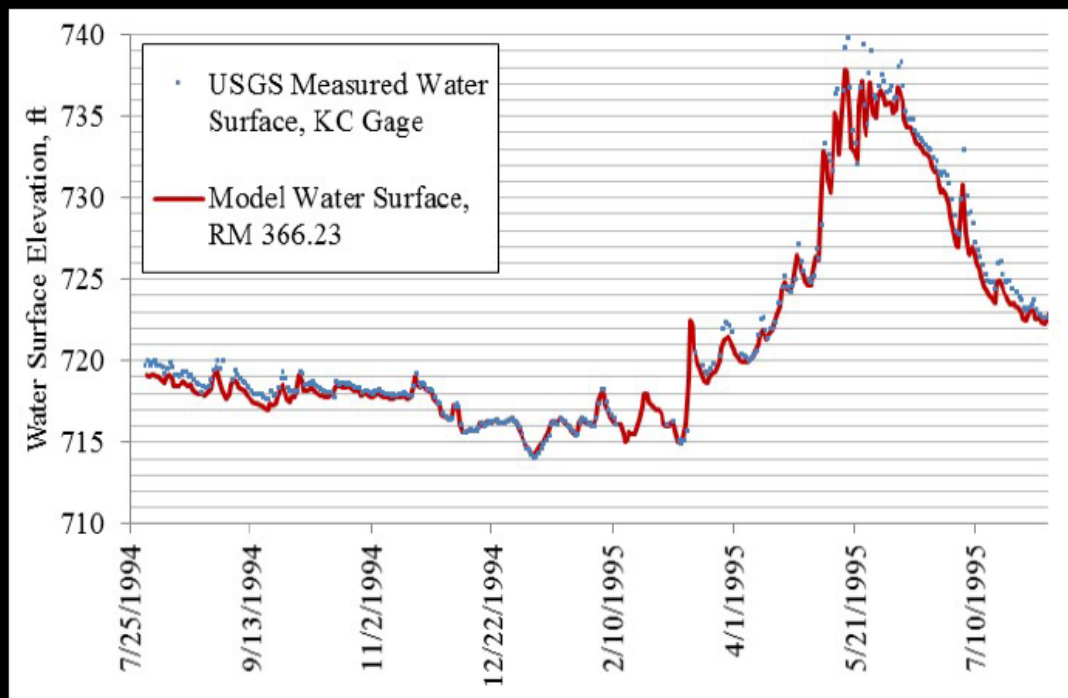
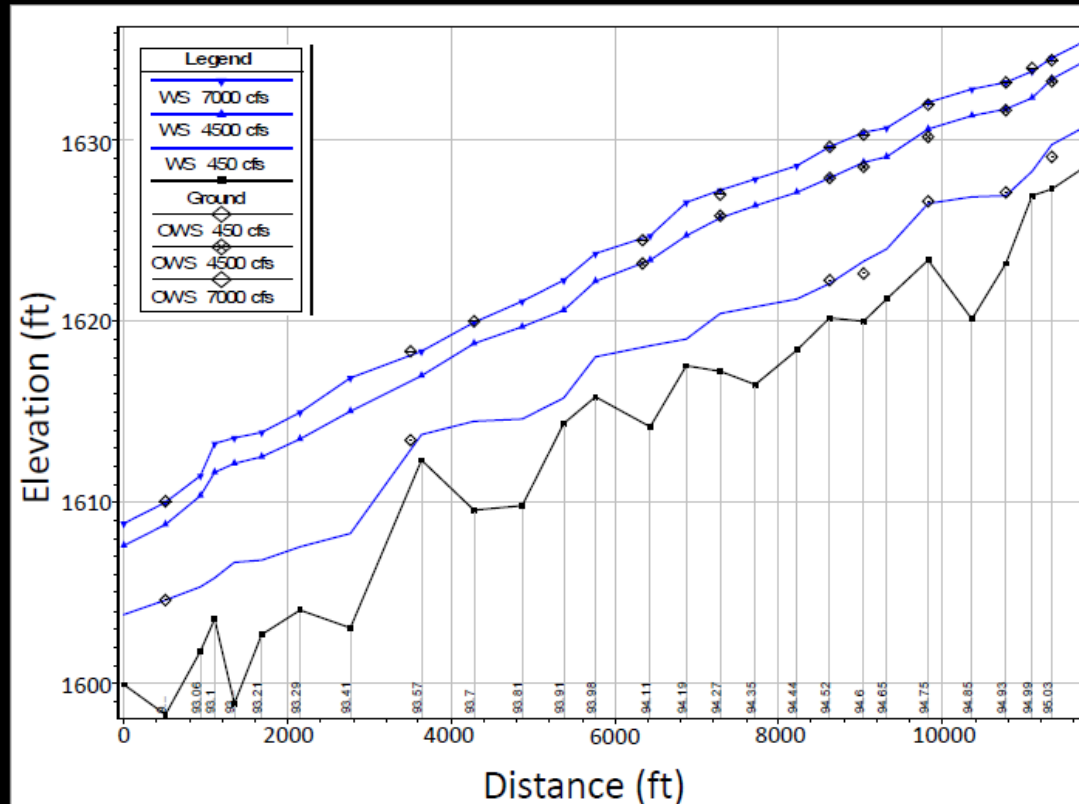


Calibration

Steady
Flow

Calibration

Unsteady
Flow



By: John Shelley, PhD (NWK)
From: Missouri River Bed Degradation Feasibility Study. USACE (2017).

Unsteady Flow: How is it different than Steady Flow?

- Boundary Conditions
- Calibration
- **Model Applications**

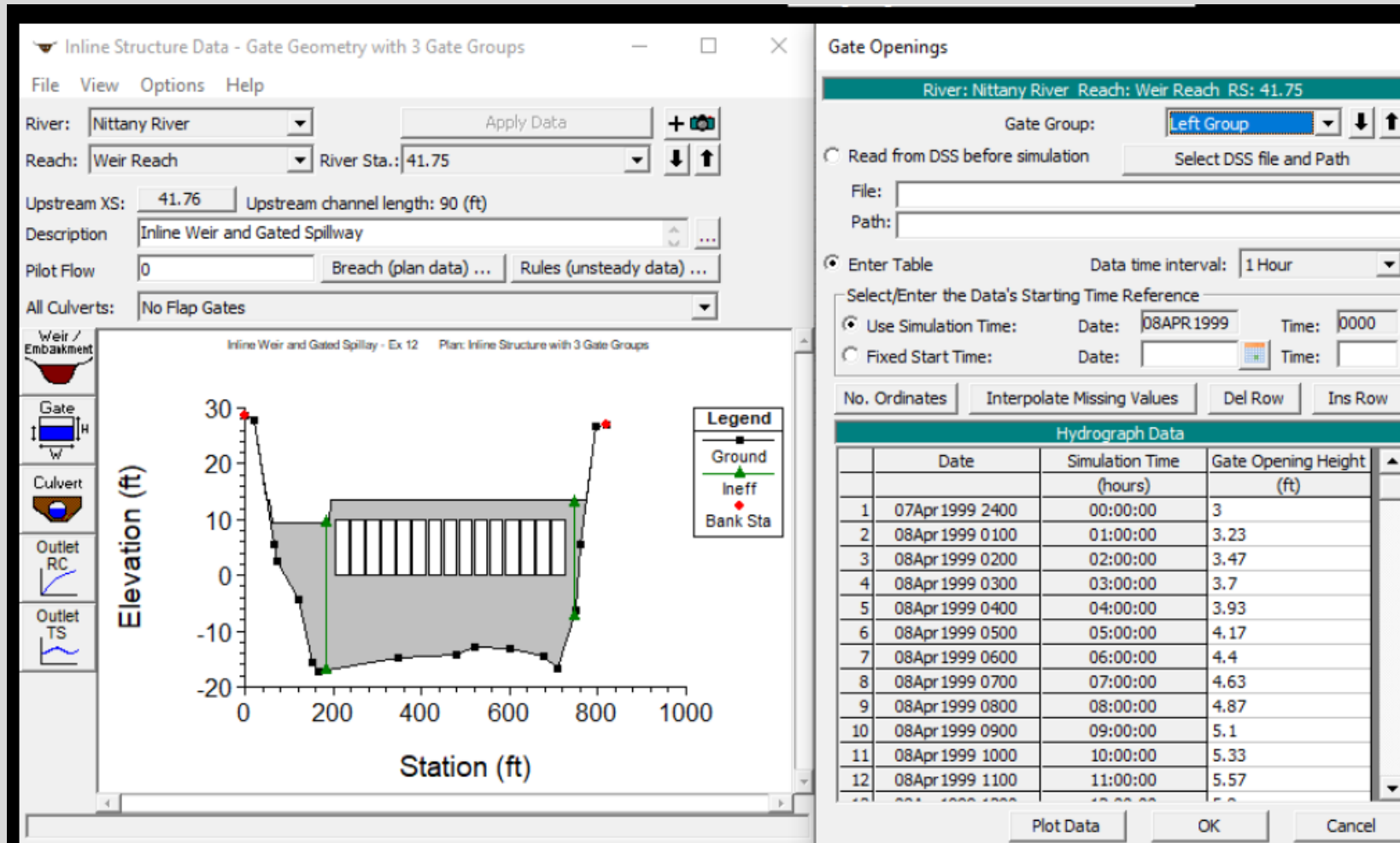
When should unsteady flow modelling be used instead of steady flow modelling?

- Ask yourself:
 - Does volume matter?
 - Does timing matter?

Are there unique aspects of the system that will significantly affect the choice of the computational model?

- River is tidally influenced
- Do wind speed and direction affect the water surface elev.
- Is the river affected by floating ice or ice jams
- Does debris tend to pile up at bridges or hydraulic structures
- Will dam/levee overtopping and breaching need to be analyzed
- Are there bridges/culverts, dams, and weirs that need to be modeled.
- Are there gated systems and pump stations that must be operated to analyze this system.

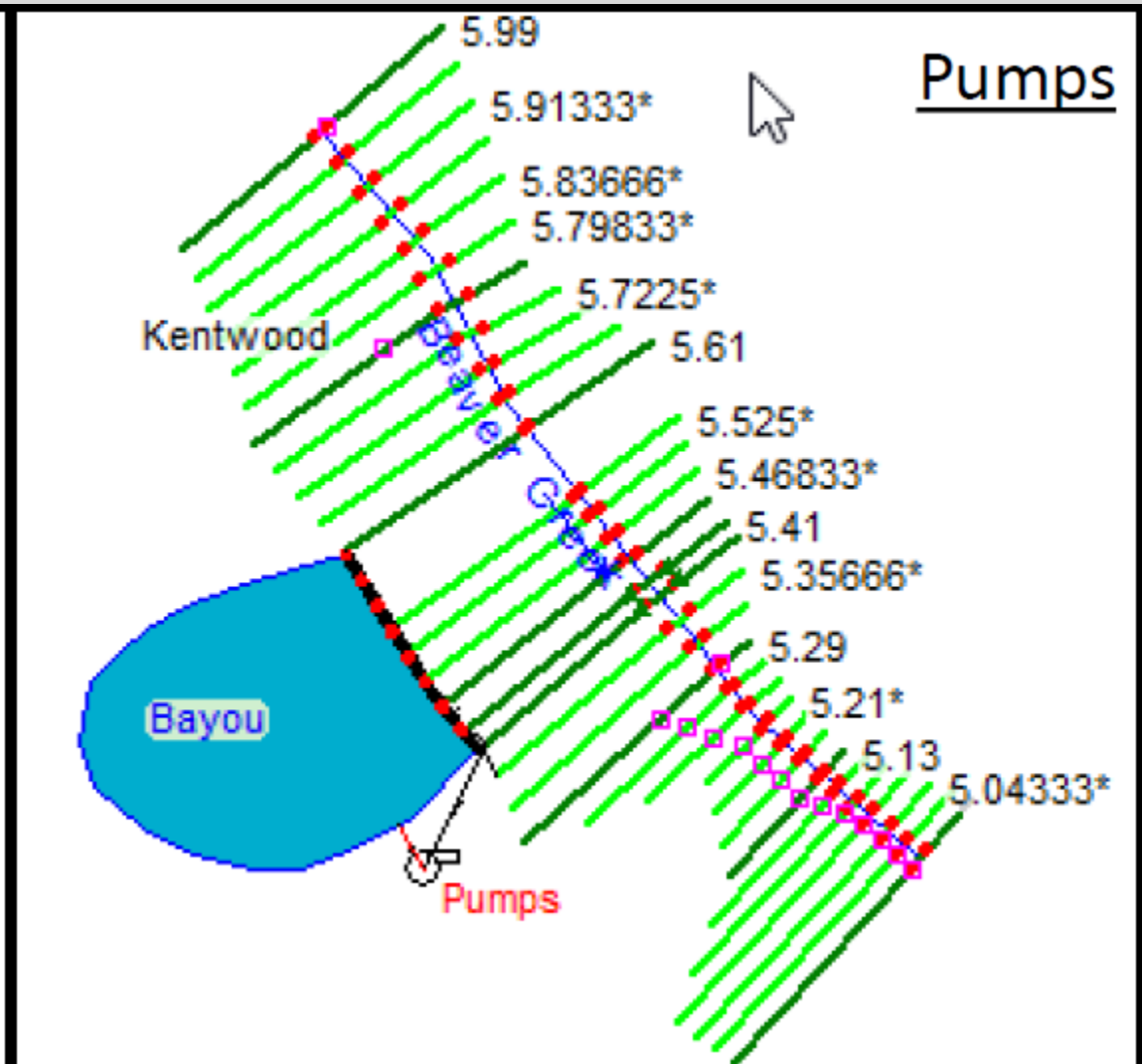
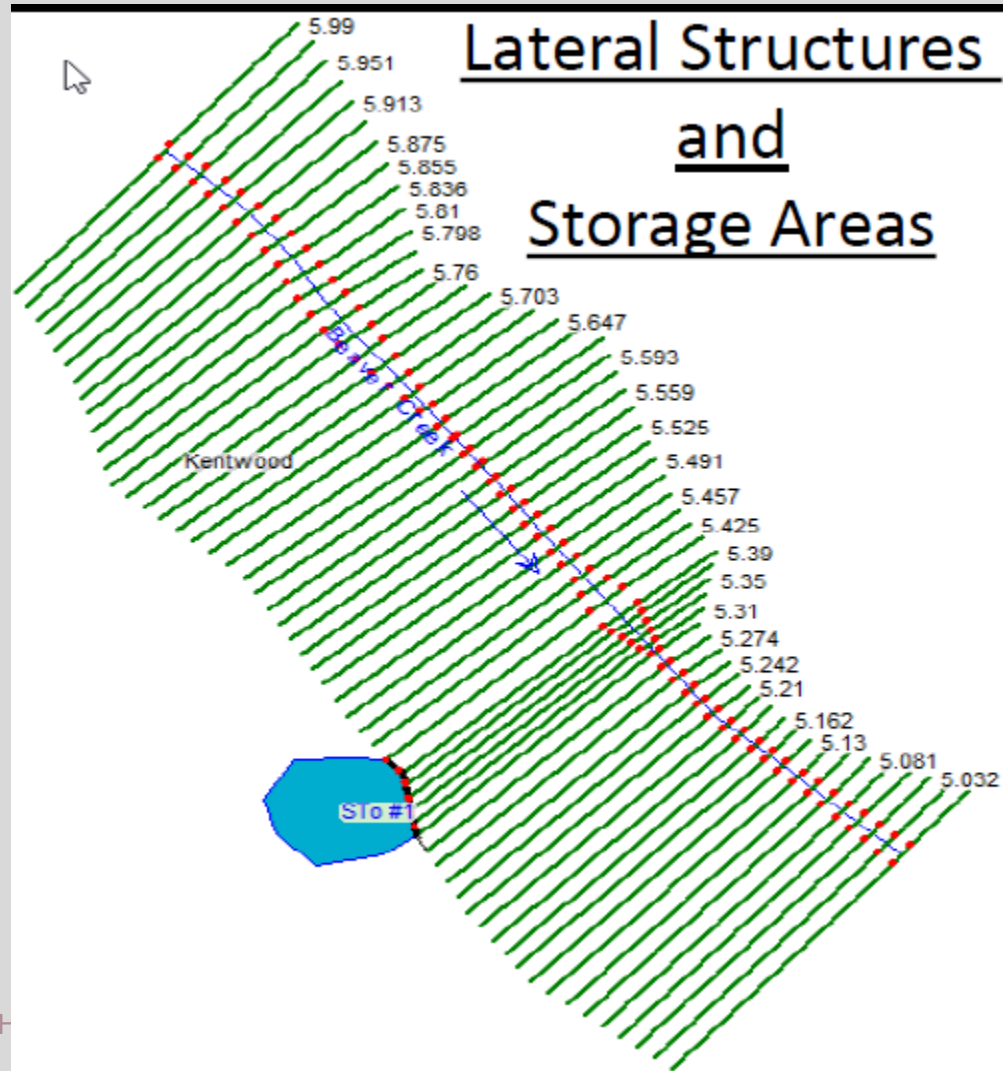
Applications: Dams and Reservoirs



Dams and Reservoirs:

- All – Volume Matters
- Operations $-f(t)$
 - Specified
 - Rules
- Dam Breach

Applications: Time sensitive diversion or attenuation storage



Applications: Operational Rules

Rule Operations

Description: Divert flow out of the system by using a two way table. Note the lateral structure is "abstract" (no weir or gate flow computations are used).

Gate Parameters				
Location	Open Rate (ft/min)	Close Rate (ft/min)	Max Opening	Min O
1				

Summary of Variable Initializations:

User Variable	Description
1	

Rule Operations

row	Operation
1	! Get the Hour of the Day
2	'Hour of Day' = Time:Hour of Day(Beginning of time step)
3	!
4	! Get the average flow at the dam over the last hour;
5	! Average flow over an hour is used to prevent sudden changes.
6	'Flow at Inline Weir' = Inline Structures:Structure.Total Flow(Nittany River,W...
7	!
8	! Based on both the Flow and the Time;
9	! Use a 2-way table to look up the flow to divert.
10	'Flow to Divert' = Table Lookup(Inline Flow, Hour, Interpolate value)
11	!
12	! Set the amount of flow to divert.
13	Structure.Total Flow (Fixed) = 'Flow to Divert'

Enter/Edit Rule Operations... OK Cancel

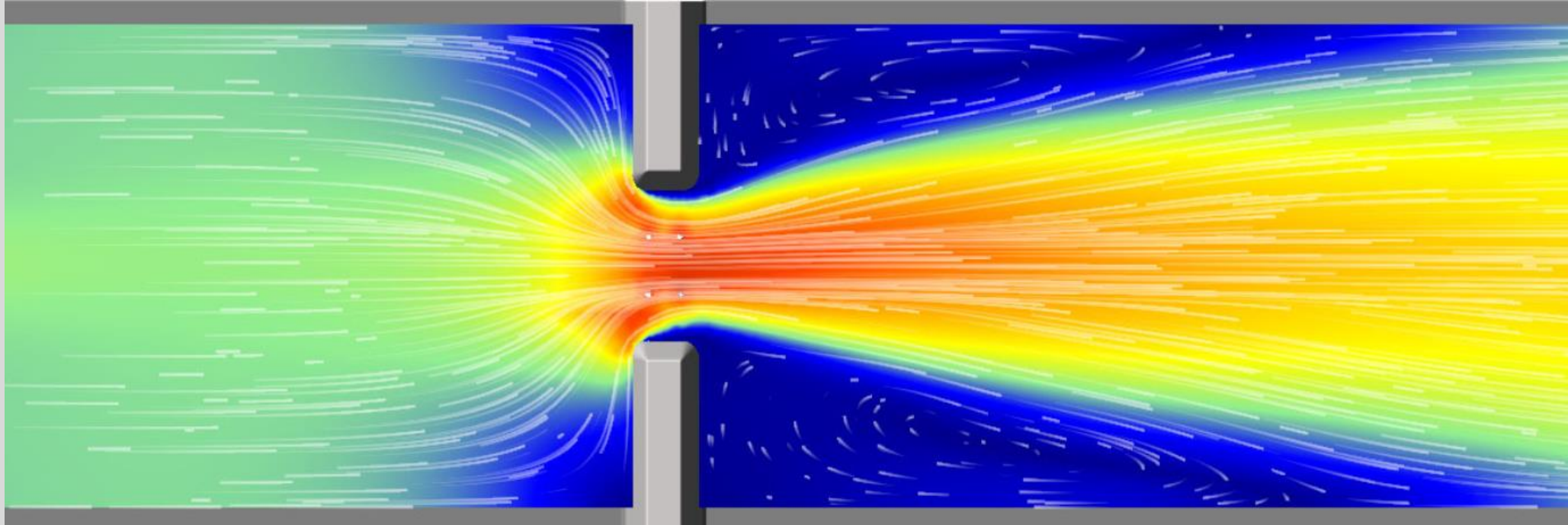
Operational Rules:

- Dam Operations
- Lateral Gates
- Pumps
- Sediment

Culvert Modelling in HEC-RAS

- See Computer Lab

Bridge Modelling in HEC-RAS



To ponder

When would you use steady or unsteady flow?

1. Big
2. Quick
3. Perks

Bridge modelling in HEC-RAS

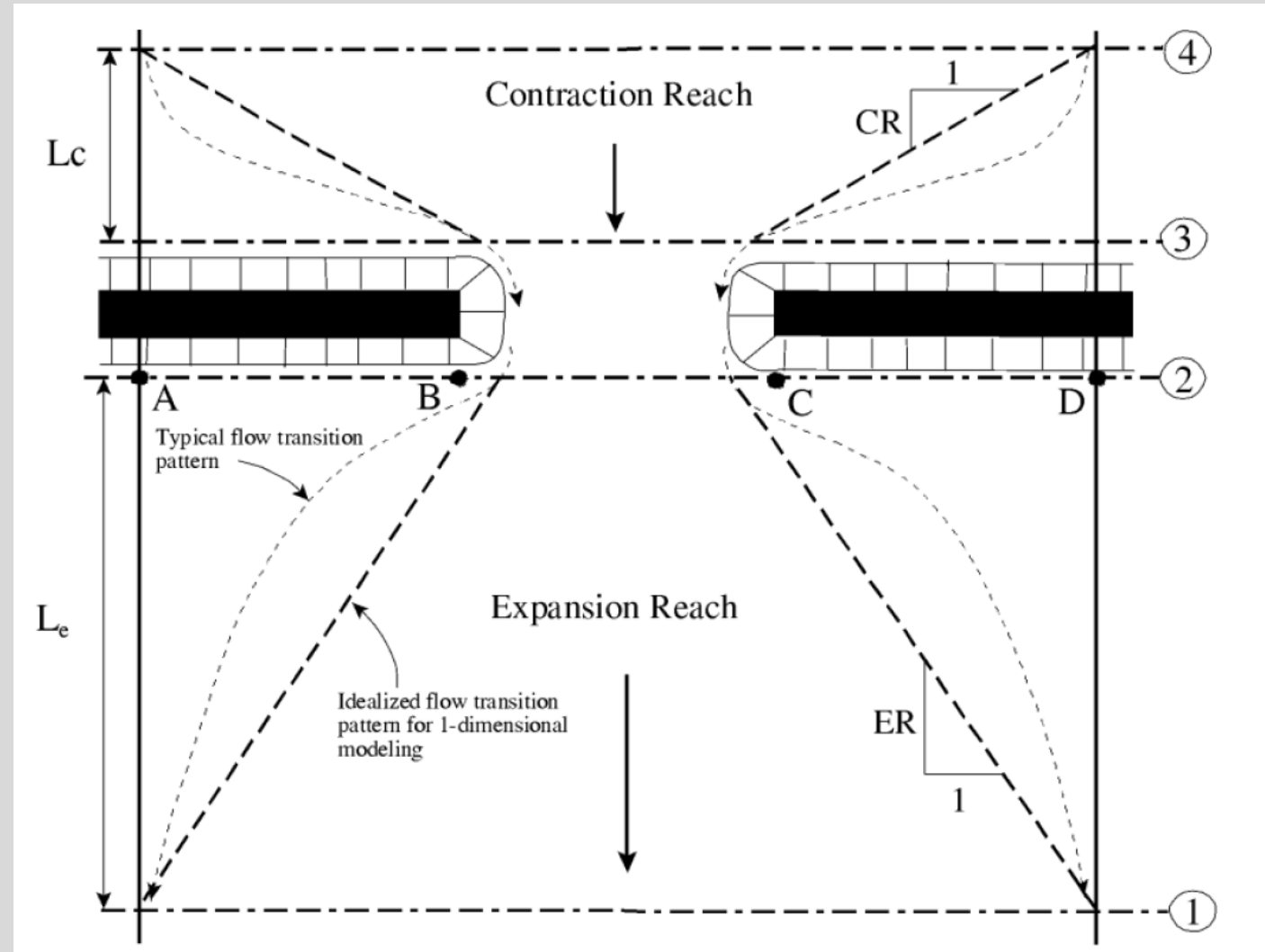
Bridge Modelling in HEC-RAS

- Energy losses computed in 3 parts:

1. in the reach immediately downstream from the structure (generally expansion of flow)
2. at the structure itself
3. in the reach immediately upstream of the structure (generally contraction of flow)

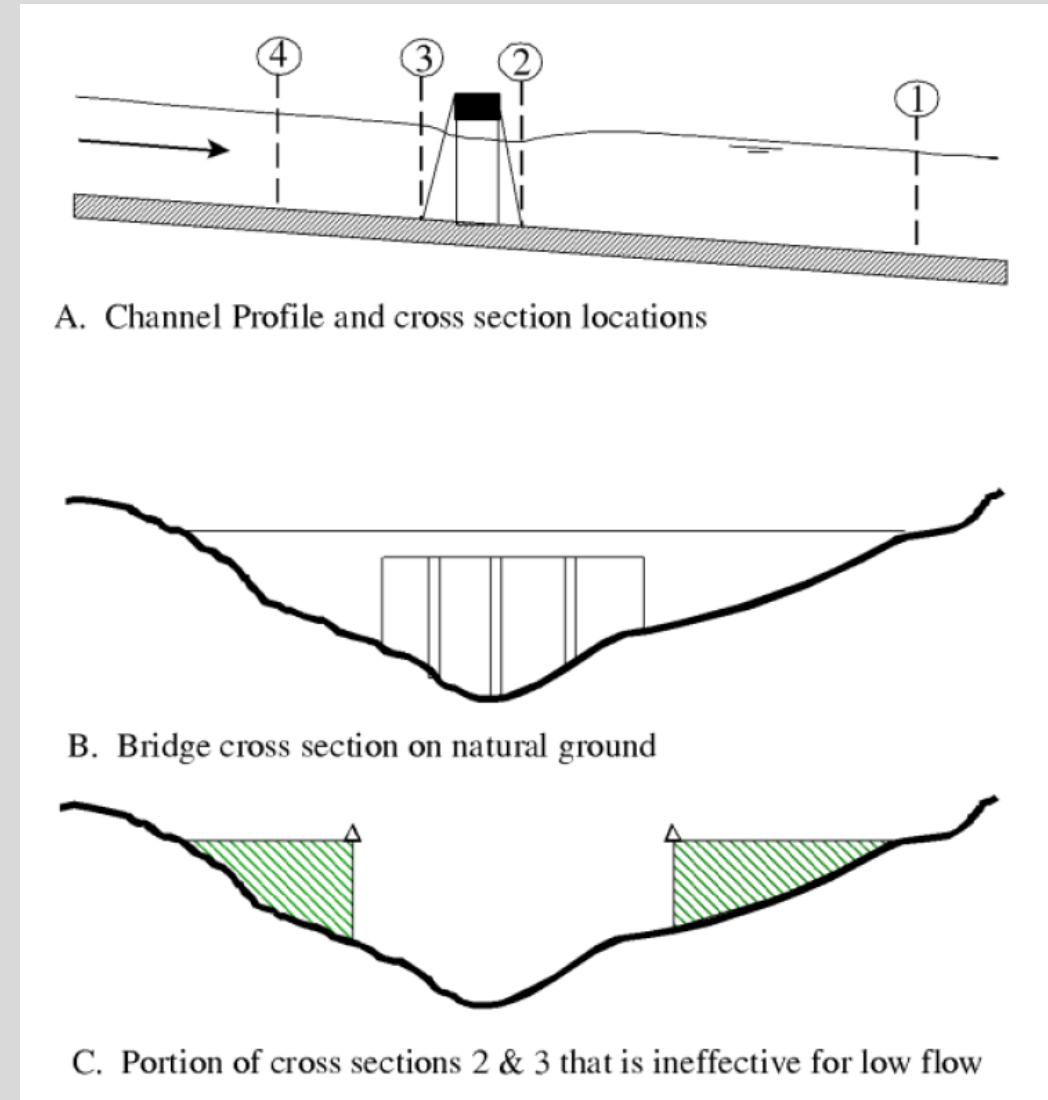
Recall contraction and expansion loss coefficients:

	Contraction	Expansion
No transition loss computed	0.0	0.0
Gradual transitions	0.1	0.3
Typical Bridge sections	0.3	0.5
Abrupt transitions	0.6	0.8



Defining Ineffective Flow Areas

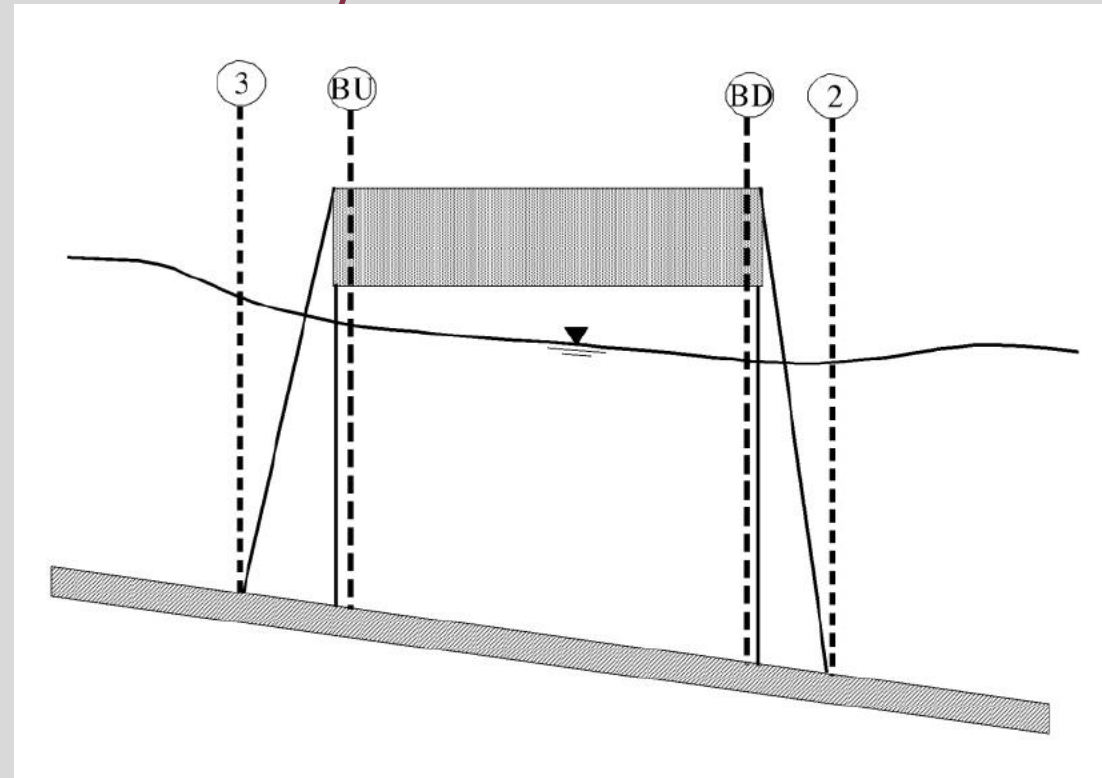
- Example: Bridge spans the entire floodway, abutments obstruct the natural floodplain.
- Active vs ineffective flow areas



Hydraulic Computations through the Bridge (here: low flow)

There are four methods available for computing losses through the bridge (sections 2 to 3):

- 1) Energy Equation (standard step method)
- 2) Momentum Balance
- 3) Yarnell Equation
- 4) FHWA WSPRO method



1) Energy Equation

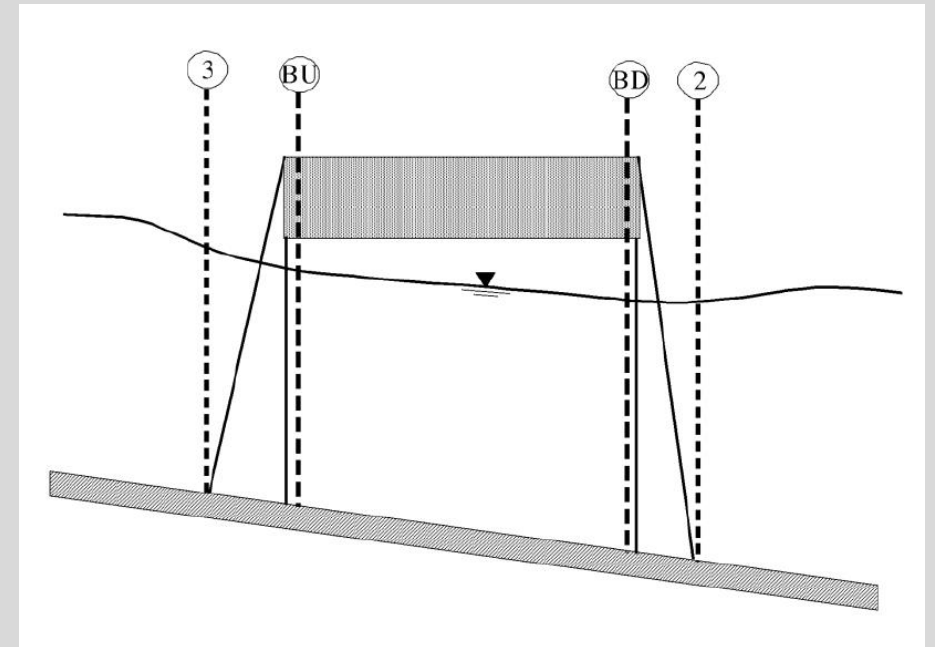
Bridge **treated in the same manner as a natural river cross-section**, except area of the bridge below the **water surface is subtracted from the total area**, and the **wetted perimeter is increased** where the water is **in contact with the bridge** structure.

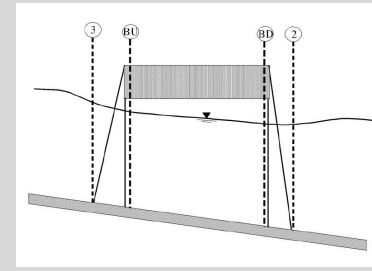
Standard step calculation sequence:

- from section 2 to section BD
- from section BD to section BU
- from section BU to section 3

Required:

- Manning's n (friction loss)
- Contraction and expansion coefficients (transition loss)





2) Momentum Balance Method

Momentum balance is performed **from cross section 2 to cross section 3**.

Three steps:

- from section 2 to section BD

$$A_{BD}\bar{Y}_{BD} + \frac{\beta_{BD}Q_{BD}^2}{gA_{BD}} = A_2\bar{Y}_2 + \frac{\beta_2Q_2^2}{gA_2} - A_{PBD}\bar{P}_{BD} + F_f - W_x$$

- from section BD to section BU

$$A_{BU}\bar{Y}_{BU} + \frac{\beta_{BU}Q_{BU}^2}{gA_{BU}} = A_{BD}\bar{Y}_{BD} + \frac{\beta_{BD}Q_{BD}^2}{gA_{BD}} + F_f - W_x$$

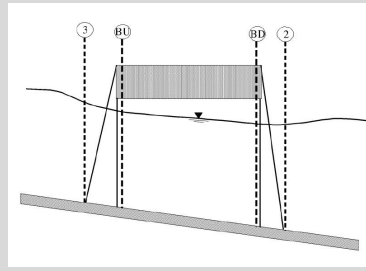
- from section BU to section 3

$$A_3\bar{Y}_3 + \frac{\beta_3Q_3^2}{gA_3} = A_{BU}\bar{Y}_{BU} + \frac{\beta_{BD}Q_{BD}^2}{gA_{BD}} + A_{PBU}\bar{Y}_{PBU} + \frac{1}{2}C_D\frac{A_{PBU}Q_3^2}{gA_3^2} + F_f - W_x$$

Required:

- Roughness coefficients (friction force)
- Drag coefficient (force of drag on piers)

3) Yarnell Equation



Empirical equation used to predict the change in water surface from just **section 2 to section 3**. The equation is **based on approximately 2600 lab experiments** in which the shape of the piers, the width, the length, the angle, and the flow rate have been varied (Yarnell, 1934):

$$H_{3-2} = 2K(K + 10\omega - 0.6)(\alpha + 15\alpha^4) \frac{V^2}{2g}$$

H_{3-2}	Drop in water surface elevation from section 3 to 2
K	Yarnell's pier shape coefficients
ω	Ratio of velocity head to depth at section 2
α	Obstructed area of the piers divided by the total unobstructed area at section 2
V^2	Velocity downstream at section 2

- The **upstream water surface** elevation (section 3) is the downstream water surface elevation plus H_{3-2} .
- **Section 3 velocity head and energy elevation** computed using upstream WS.

Required:

- Pier shape coefficient K

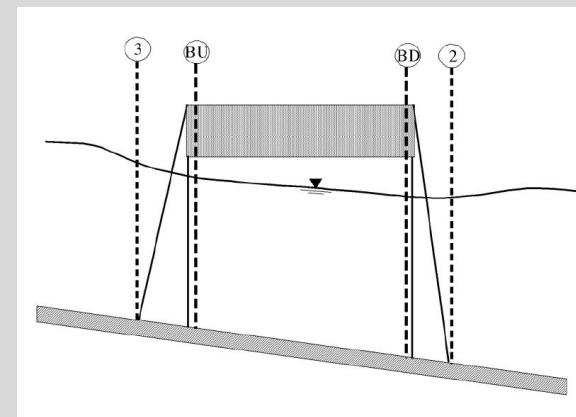
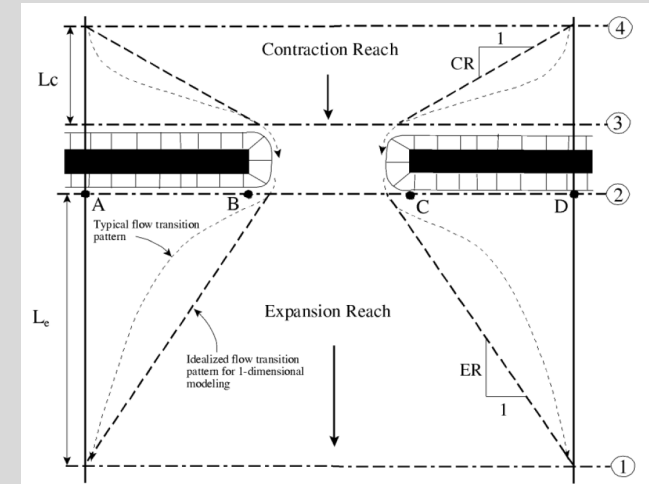
4) FHWA WSPRO method

Solves the energy equation in an iterative manner, from the exit cross section (1) to the approach cross-section (4).

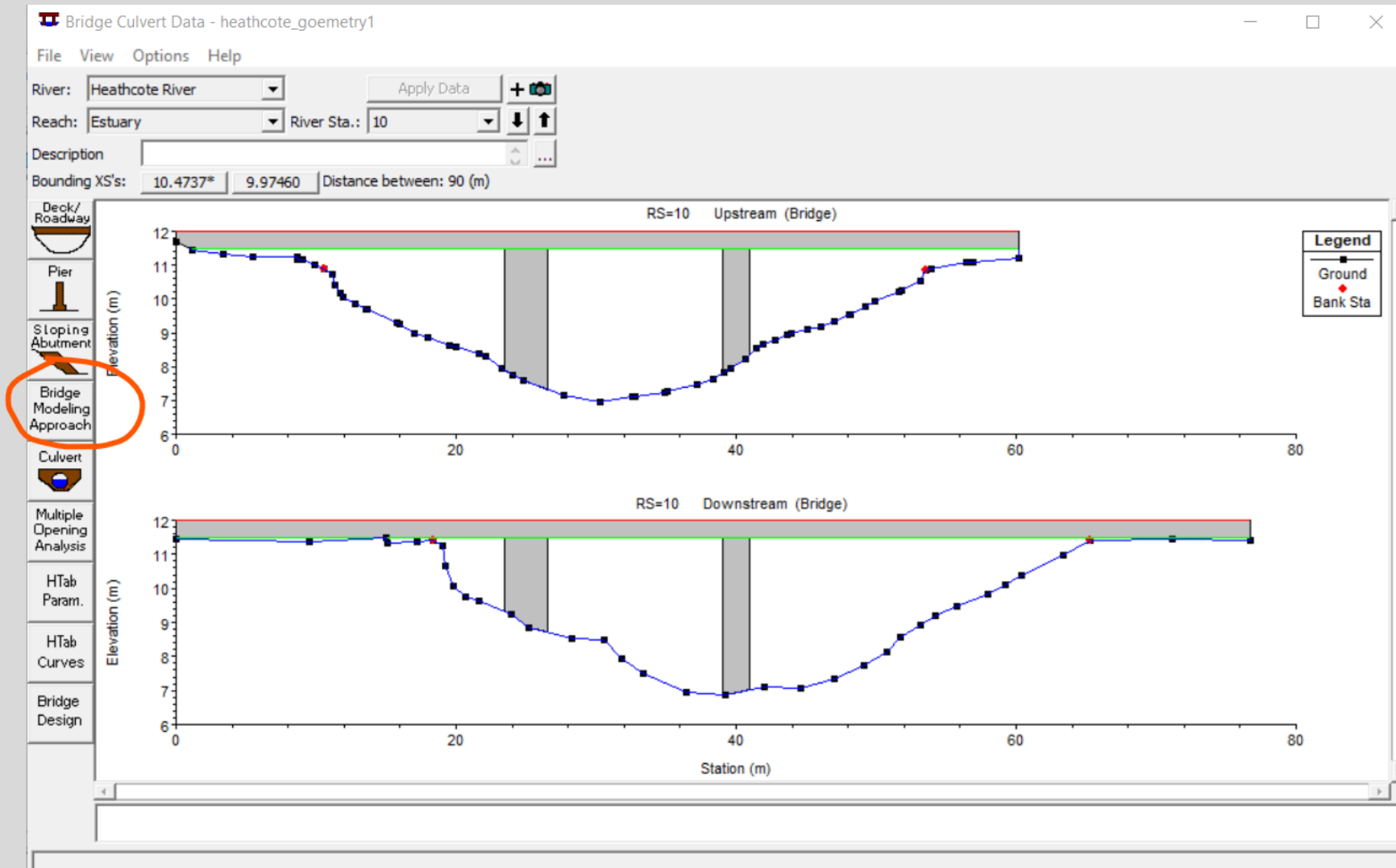
$$h_4 + \frac{a_4 V_4^2}{2g} = h_1 + \frac{a_1 V_1^2}{2g} + h_f + h_e$$

The energy balance is performed in the following steps

- from section (1) to section (2);
- from section (2) to section (BD);
- from section (BD) to section (BU);
- from section (BU) to section (3); and
- from section (3) to section (4).



Option selection in HEC-RAS



Option selection in HEC-RAS

Bridge Modeling Approach Editor

Add Copy Delete Bridge # 1 ↓ ↑

Low Flow Methods

Use Compute

☐ ☒ Energy (Standard Step)

☐ ☒ Momentum Coef Drag Cd 1 ?

☐ ☒ Yarnell (Class A only) Pier Shape K 1 ?

☐ ☐ WSPRO Method (Class A only) WSPRO Variables

☒ Highest Energy Answer

High Flow Methods

☒ Energy Only (Standard Step)

☐ Pressure and/or Weir

Submerged Inlet Cd (Blank for table)

Submerged Inlet + Outlet Cd 0.8

Max Low Chord (Blank for default)

OK Cancel Help

Enter to add another bridge coefficient set.